

Headlight Restoration Employee Training Guide

Employee Training • Customer Education • Professional Detailing Standards

Purpose of this guide

This guide gives employees a consistent way to assess headlight condition, perform safe restoration, and set accurate customer expectations about how long results last.

CORE PRINCIPLE

***"Restoration renews the surface.
It doesn't fix what's happening inside the housing."***

Origination: Generated by Claude (Anthropic) for Prime Auto Detail, September 17, 2026

1. Why Headlights Oxidize

Most modern headlight lenses are polycarbonate plastic with a factory UV-protective clear coat. Over time, sun exposure breaks down that clear coat, allowing the polycarbonate underneath to oxidize — causing the hazy, yellowed appearance customers ask about.

2. Assessing Headlight Condition

Condition	Restoration Candidate?
Light surface haze	Yes — often polish-only, no sanding needed
Moderate-to-deep yellowing	Yes — full wet-sand progression needed
Cracked lens	No — restoration won't fix structural damage; recommend replacement
Foggy/moisture inside the housing	No — indicates a seal failure; polishing the outside won't help; recommend replacement or seal repair

Employee rule: Always assess before quoting. Selling restoration on a lens that's cracked or fogged internally sets the customer up for disappointment and reflects badly on the shop.

3. Tools & Materials

[PLACEHOLDER: insert your actual wet-sand grit progression and product names here]

- Progressive wet-sand grits (example progression: coarse → medium → fine, commonly in the 800–3000 range) — confirm your actual grit set.
- Buffing/polishing compound for post-sand clarity.
- UV-protective sealant or coating — this step is not optional; skipping it means the oxidation returns within weeks.
- Painter's tape/masking to protect surrounding paint and trim from sanding debris and compound.

4. Step-by-Step Restoration Process

- Mask off all surrounding paint, trim, and badges — sanding debris and compound can dull or scratch adjacent paint if not protected.
- Wet sand progressively through each grit, from coarsest to finest, keeping the surface wet throughout to avoid heat buildup and deeper scratching. Do not skip a grit step — each one is there to remove the scratch pattern left by the previous one.
- Once sanding is complete and the surface is uniformly hazy/matte (not clear yet), compound and polish the lens to restore clarity.
- Clean and fully dry the lens.
- Apply the UV-protective coating per the product's exact cure instructions — this is what prevents oxidation from returning quickly.

5. Scenario Guidance

- **Mild haze only:** May not need the full sanding progression — a polish-only pass can sometimes achieve clarity, saving time. Assess before defaulting to full sanding.
- **Deep, heavy yellowing:** Requires the full grit progression — skipping early grits to save time leaves visible scratch patterns under the final polish.
- **OEM factory-coated lenses vs. aftermarket replacements:** Aftermarket lenses may have different coating properties — if results seem inconsistent with your usual process, note it rather than assuming technique error.

6. Common Mistakes to Call Out

- Skipping a grit step in the sanding progression — leaves visible swirl/scratch patterns under the final polish that will show clearly once headlights are on at night.
- Insufficient masking, letting sanding slurry or compound land on and dull surrounding paint.
- Skipping or rushing the UV coating application — this is the single biggest reason a "restored" headlight looks hazy again within a month.
- Letting the customer drive off before the UV coating's cure time has been met.

7. Customer Expectations & Script

Employee rule: Restoration is a renewable maintenance service, not a permanent fix — set the re-coat interval expectation up front.

"Headlight restoration removes the oxidation and haze and restores clarity, then we seal it with a UV-protective coating to slow that oxidation from coming back. It's not a one-time-forever fix — direct sun exposure will start the process again over time, so we'd recommend a maintenance re-coat down the road to keep them clear."

Quick Reference Rules

- Always assess for cracks or internal fogging before quoting — these aren't restoration candidates.
 - Always mask surrounding paint/trim before sanding.
 - Never skip a grit step in the sanding progression.
 - Always apply the UV coating — never skip this step.
 - Always respect the coating's cure time before customer pickup.
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